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Roll No

EE-304 (GS)

B.E. III Semester Examination, June 2020

Grading System (GS)

Semiconductor Devices and Circuits

Time : Three Hours

Maximum Marks : 70

Note: i) Answer any five questions.

ii) All questions carry equal marks.

1. Explain photo diode with respect to following points:
 - i) Construction
 - ii) Operation
 - ii) Characteristics
 - iv) Applications
2. Calculate the frequency of oscillations for Wien bridge oscillation circuit with $R = 52 \text{ k}\Omega$ and $C = 0.0011 \mu\text{F}$.
3. Name different types of Oscillator? Explain one of them with circuit diagram and working.
4. A common base transistor amplifier is driven by a voltage source V_s and internal resistance $R_h = 1200 \Omega$. The load impedance is a resistor R_L of 1000Ω . The 'h' parameters are given below:
 $h_{ib} = 220 \Omega$ $h_{rb} = 3 \times 10^{-4}$
 $h_{fb} = -0.98$ $h_{ob} = 0.5 \mu\text{A/V}$
Compute, current gain (A_i), input impedance (R_i), Voltage gain (A_v), input impedance (R_o) and power gain (A_p)
5. With the help of circuit diagram explain the working of 555 timer. Also discuss its applications.
6. Sketch and explain the followings for a transistor :
 - i) CB output characteristic indicating all the regions.
 - ii) CE input characteristic.

OR

Explain the working of N channel enhancement MOSFET. Also draw and explain the transfer characteristic curves.

7. Explain the formation of the depletion layer in an unbiased pn-junction. How does it vary with external bias?

OR

What is a Bistable Circuit? Discuss the operation of a transistor bistable multivibrator.

8. Write short note on any two of the following:
 - a) Common Mode Rejection Ratio (CMRR)
 - b) Slew rate
 - c) Schmitt trigger
